1. Load the given retail sales dataset into a Pandas DataFrame and display the first 5 rows along with the dataset information to understand its structure.

```python
import pandas as pd

df = pd.read_csv(r"C:\Users\amrit\OneDrive\Desktop\PW Python\retail_dataset .csv")

print("--- First 5 Rows of the Retail Dataset ---")
print(df.head())
```

```
it/OneDrive/Desktop/PW Python/assign.py"
--- First 5 Rows of the Retail Dataset ---
  Transaction ID Customer ID Product Category  Quantity   Price        Date Region
0           T001        C015            Books       7.0   167.0  02-02-2023  North
1           T002        C004             Toys       4.0   374.0  17-02-2023  South
2           T003        C033        Furniture       8.0  1059.0  17-05-2024  South
3           T004        C047        Groceries       5.0   304.0  11-02-2024  South
4           T005        C022            Books       7.0   282.0  27-07-2024   West
PS C:\Users\amrit\OneDrive\Desktop\PW Python> (C:\Users\amrit\anaconda3\shell\condabin\cond
```

2. Inspect the dataset and identify missing values in each column. Display the total number of missing values present in the dataset.

```python
import pandas as pd

# Load the dataset
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW Python\retail_dataset .csv')

# 1. Identify missing values in each column
missing_per_column = df.isnull().sum()
print("--- MISSING VALUES PER COLUMN ---")
print(missing_per_column)

# 2. Calculate and display the total number of missing values in the entire dataset
total_missing = df.isnull().sum().sum()
print("\n--- TOTAL MISSING VALUES IN DATASET ---")
print(f"Total missing values: {total_missing}")
```

```
--- MISSING VALUES PER COLUMN ---
Transaction ID      0
Customer ID         0
Product Category    3
Quantity            2
Price               5
Date                0
Region              2
dtype: int64

--- TOTAL MISSING VALUES IN DATASET ---
Total missing values: 12
```

3. Replace missing values in the Quantity and Price columns with their respective mean values using Pandas.

```
import pandas as pd

# Load the dataset
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')

quantity_mean = df['Quantity'].mean()
price_mean = df['Price'].mean()

print("--- CALCULATED MEANS ---")
print(f"Mean Quantity: {quantity_mean:.2f}")
print(f"Mean Price: ${price_mean:.2f}\n")

# 2. Replace missing values using fillna()
df['Quantity'] = df['Quantity'].fillna(quantity_mean)
df['Price'] = df['Price'].fillna(price_mean)

# Verify the changes by checking missing values in these columns
print("--- MISSING VALUES AFTER IMPUTATION ---")
print(df[['Quantity', 'Price']].isnull().sum())
```

```
--- CALCULATED MEANS ---
Mean Quantity: 5.96
Mean Price: $820.53

--- MISSING VALUES AFTER IMPUTATION ---
Quantity    0
Price       0
dtype: int64
```

4. Remove all rows where the Product Category or Region column contains missing values.

```
import pandas as pd

# Load the dataset
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')

# Record the initial number of rows
initial_rows = df.shape[0]

# Remove rows where 'Product Category' or 'Region' contains missing
values
# subset=['Product Category', 'Region'] isolates the check to just
these columns
df_cleaned = df.dropna(subset=['Product Category', 'Region'])

# Display the result and rows removed
print(f"Initial row count: {initial_rows}")
print(f"Row count after dropping missing categories/regions:
{df_cleaned.shape[0]}")
print(f"Total rows dropped: {initial_rows - df_cleaned.shape[0]}")
```

```
Initial row count: 100
Row count after dropping missing categories/regions: 95
Total rows dropped: 5
PS C:\Users\amrit\OneDrive\Desktop\PW Python> (C:\Users\
```

5. Create a new column called Revenue using NumPy, where:
   Revenue = Quantity × Price
   Display the updated dataset with the new column.

```
import pandas as pd
import numpy as np

# 1. Load and prep dataset based on previous steps
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')

# Impute missing numeric columns (from Q3)
df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
df['Price'] = df['Price'].fillna(df['Price'].mean())

# Drop missing categorical rows (from Q4)
df_cleaned = df.dropna(subset=['Product Category', 'Region']).copy()

# 2. Calculate Revenue using NumPy vectorization
# Converting columns to NumPy arrays explicitly ensures NumPy
handles the operation
df_cleaned['Revenue'] =
np.multiply(df_cleaned['Quantity'].to_numpy(),
df_cleaned['Price'].to_numpy())

# 3. Display the first 5 rows of the updated dataset
print("--- UPDATED DATASET WITH REVENUE COLUMN (FIRST 5 ROWS) ---")
print(df_cleaned[['Transaction ID', 'Product Category', 'Quantity',
'Price', 'Revenue', 'Region']].head())
```

```
--- UPDATED DATASET WITH REVENUE COLUMN (FIRST 5 ROWS) ---
  Transaction ID Product Category  Quantity   Price  Revenue Region
0           T001            Books       7.0   167.0   1169.0  North
1           T002             Toys       4.0   374.0   1496.0  South
2           T003        Furniture       8.0  1059.0   8472.0  South
3           T004        Groceries       5.0   304.0   1520.0  South
4           T005            Books       7.0   282.0   1974.0   West
(base) PS C:\Users\amrit\OneDrive\Desktop\PW Python>
```

6. Using NumPy, calculate the total revenue generated from all transactions in the dataset.

```
import pandas as pd
import numpy as np

# 1. Load and prep dataset based on previous steps
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')

df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
df['Price'] = df['Price'].fillna(df['Price'].mean())
df_cleaned = df.dropna(subset=['Product Category', 'Region']).copy()

# 2. Compute the Revenue column (from Q5)
df_cleaned['Revenue'] =
np.multiply(df_cleaned['Quantity'].to_numpy(),
df_cleaned['Price'].to_numpy())

# 3. Use NumPy to calculate the grand total revenue
total_revenue = np.sum(df_cleaned['Revenue'].to_numpy())

print("--- GLOBAL REVENUE REPORT ---")
print(f"Total Revenue Generated: ${total_revenue:,.2f}")
```

```
--- GLOBAL REVENUE REPORT ---
Total Revenue Generated: $465,529.66
```

7. Group the dataset by Product Category and calculate the total revenue generated by each category.

```
import pandas as pd
import numpy as np

# 1. Load and prep dataset based on previous steps
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')

df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
df['Price'] = df['Price'].fillna(df['Price'].mean())
```

```python
df_cleaned = df.dropna(subset=['Product Category', 'Region']).copy()

# 2. Compute the Revenue column
df_cleaned['Revenue'] =
np.multiply(df_cleaned['Quantity'].to_numpy(),
df_cleaned['Price'].to_numpy())

# 3. Group by Product Category and calculate total revenue
category_revenue = df_cleaned.groupby('Product
Category')['Revenue'].sum().reset_index()

# Rename columns for clarity and display the result
category_revenue.columns = ['Product Category', 'Total Revenue']
print("--- TOTAL REVENUE BY PRODUCT CATEGORY ---")
print(category_revenue.to_string(index=False, formatters={'Total
Revenue': '{:,.2f}'.format}))
```

```
--- TOTAL REVENUE BY PRODUCT CATEGORY ---
Product Category Total Revenue
          Beauty     27,647.00
           Books     29,597.94
        Clothing     85,006.00
     Electronics     93,204.32
       Furniture    129,664.53
       Groceries     11,978.00
          Sports     34,484.00
            Toys     53,947.88
```

8. Based on the revenue calculated for each product category, identify the top 3 categories with the highest revenue and the bottom 3 categories with the lowest revenue.

```python
import pandas as pd
import numpy as np

# 1. Load and prep dataset based on previous steps
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')
```

```python
df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
df['Price'] = df['Price'].fillna(df['Price'].mean())
df_cleaned = df.dropna(subset=['Product Category', 'Region']).copy()
df_cleaned['Revenue'] =
np.multiply(df_cleaned['Quantity'].to_numpy(),
df_cleaned['Price'].to_numpy())

# 2. Aggregate revenue by category
category_revenue = df_cleaned.groupby('Product
Category')['Revenue'].sum().reset_index()

# 3. Sort the categories by revenue in descending order
category_sorted = category_revenue.sort_values(by='Revenue',
ascending=False)

# 4. Extract Top 3 and Bottom 3 using head() and tail()
top_3 = category_sorted.head(3)
bottom_3 = category_sorted.tail(3)

print("--- TOP 3 HIGHEST REVENUE CATEGORIES ---")
print(top_3.to_string(index=False, formatters={'Revenue':
'${:,.2f}'.format}))

print("\n--- BOTTOM 3 LOWEST REVENUE CATEGORIES ---")
# Reversing the order of tail to show lowest first for clear reading
print(bottom_3.iloc[::-1].to_string(index=False,
formatters={'Revenue': '${:,.2f}'.format}))
```

```
--- TOP 3 HIGHEST REVENUE CATEGORIES ---
Product Category     Revenue
       Furniture $129,664.53
     Electronics  $93,204.32
        Clothing  $85,006.00

--- BOTTOM 3 LOWEST REVENUE CATEGORIES ---
Product Category    Revenue
       Groceries $11,978.00
          Beauty $27,647.00
           Books $29,597.94
(base) PS C:\Users\amrit\OneDrive\Desktop\PW P
```

9. Group the dataset by Region and calculate the total revenue generated in each region. Identify the region with the highest revenue and the region with the lowest revenue.

```
import pandas as pd
import numpy as np

# 1. Load, prep, and calculate revenue (carrying over steps)
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')
df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
df['Price'] = df['Price'].fillna(df['Price'].mean())
df_cleaned = df.dropna(subset=['Product Category', 'Region']).copy()
df_cleaned['Revenue'] =
np.multiply(df_cleaned['Quantity'].to_numpy(),
df_cleaned['Price'].to_numpy())

# 2. Group by Region and calculate total revenue
region_revenue = df_cleaned.groupby('Region')['Revenue'].sum()

print("--- TOTAL REVENUE BY REGION ---")
for region, rev in region_revenue.items():
    print(f"{region:<10} : ${rev:,.2f}")

# 3. Identify the highest and lowest regions
```

```
highest_region = region_revenue.idxmax()
lowest_region = region_revenue.idxmin()

print("\n--- PERFORMANCE SUMMARY ---")
print(f"Region with HIGHEST Revenue: {highest_region}
(${region_revenue[highest_region]:,.2f})")
print(f"Region with LOWEST Revenue : {lowest_region}
(${region_revenue[lowest_region]:,.2f})")
```

```
python.exe "c:/Users/amrit/OneDrive/Desktop/PW Pytho
--- TOTAL REVENUE BY REGION ---
East         : $126,232.04
North        : $153,244.35
South        : $104,855.74
West         : $81,197.53

--- PERFORMANCE SUMMARY ---
Region with HIGHEST Revenue: North ($153,244.35)
Region with LOWEST Revenue : West ($81,197.53)
(base) PS C:\Users\amrit\OneDrive\Desktop\PW Python>
```

10. Extract the month from the Date column using Pandas and group the dataset by month to calculate the total monthly revenue. Also, calculate the mean, median, and standard deviation of the Revenue column using NumPy.

```
import pandas as pd
import numpy as np

# 1. Load, prep, and calculate revenue (carrying over steps)
df = pd.read_csv(r'C:\Users\amrit\OneDrive\Desktop\PW
Python\retail_dataset .csv')
df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
df['Price'] = df['Price'].fillna(df['Price'].mean())
df_cleaned = df.dropna(subset=['Product Category',
'Region']).copy()
df_cleaned['Revenue'] =
np.multiply(df_cleaned['Quantity'].to_numpy(),
df_cleaned['Price'].to_numpy())
```

```
# 2. Convert Date to datetime format and extract the month
# Using dayfirst=True since the dates in the CSV are formatted as
DD-MM-YYYY
df_cleaned['Date'] = pd.to_datetime(df_cleaned['Date'],
dayfirst=True)
df_cleaned['Month'] = df_cleaned['Date'].dt.month

# 3. Group by month and calculate total monthly revenue
monthly_revenue =
df_cleaned.groupby('Month')['Revenue'].sum().reset_index()
monthly_revenue.columns = ['Month', 'Total Revenue']

print("--- TOTAL REVENUE BY MONTH ---")
# Optional: Map month numbers to short names for better readability
month_names = {1:'Jan', 2:'Feb', 3:'Mar', 4:'Apr', 5:'May',
6:'Jun',
               7:'Jul', 8:'Aug', 9:'Sep', 10:'Oct', 11:'Nov',
12:'Dec'}
monthly_revenue_display = monthly_revenue.copy()
monthly_revenue_display['Month'] =
monthly_revenue_display['Month'].map(month_names)

print(monthly_revenue_display.to_string(index=False,
formatters={'Total Revenue': '${:,.2f}'.format}))

# 4. Calculate statistical metrics using NumPy
revenue_array = df_cleaned['Revenue'].to_numpy()

rev_mean = np.mean(revenue_array)
rev_median = np.median(revenue_array)
rev_std = np.std(revenue_array)

print("\n--- REVENUE STATISTICAL METRICS ---")
print(f"Mean (Average) Revenue : ${rev_mean:,.2f}")
print(f"Median Revenue         : ${rev_median:,.2f}")
print(f"Standard Deviation     : ${rev_std:,.2f}")
```

```
--- TOTAL REVENUE BY MONTH ---
Month Total Revenue
  Jan    $43,994.00
  Feb    $33,028.00
  Mar    $10,557.00
  Apr    $57,298.53
  May    $27,915.00
  Jun    $28,190.00
  Jul    $47,851.29
  Aug    $76,958.00
  Sep    $30,719.11
  Oct    $38,017.00
  Nov    $27,662.74
  Dec    $43,339.00

--- REVENUE STATISTICAL METRICS ---
Mean (Average) Revenue : $4,900.31
Median Revenue         : $3,282.11
Standard Deviation     : $4,721.17
(base) PS C:\Users\amrit\OneDrive\Desktop\PW Py
```